

COMP3331/9331

Computer Networks and Applications

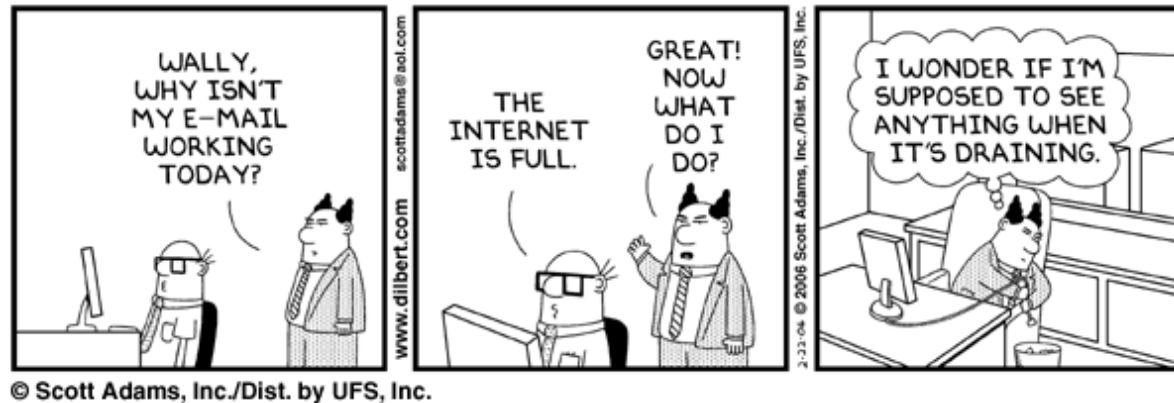
2026 Term 3

Lecturer in Charge: **Wen Hu**

Course Admin: **Isura Bamunusinghe**

Course Outline & Logistics

*1st Course on
Computer Networks*



WHY THIS MATTERS

Who cares about computer networking?

Companies whose entire business model runs on the network - 2025 revenue:

\$60B

2025 revenue

Cisco

\$46B

2025 revenue

Netflix

Every one of them is, at heart, a networking company.

2023 Optus outage

Article [Talk](#)

From Wikipedia, the free encyclopedia

The **2023 Optus outage** was an unplanned [outage](#) of all [Optus](#) internet, [cellular](#) and [fixed-line](#) services in Australia first detected on 8 November 2023 at 04:05 [AEDT](#).^[2] The outage caused disruption across government and corporate sectors including hospitals, banks, train services, [EFTPOS](#) payment systems, and calls to [emergency services](#).^{[2][3]} Restoration was gradual with some services being restored around 13:00 [AEDT](#).^[4] The outage directly affected more than 10 million people and 400,000 businesses across [Australia](#).^{[5][6]}

The cause was likely due to several routers disconnecting as an automated protective measure against routing update overload.^[1] Optus stated that the outage was not due to a [cyberattack](#).^{[2][7]}

vision language models venture



AI Mode All Images Videos **News** Shopping Maps More Tools

VB VentureBeat

Cohere Parse 5 loses the benchmark on points. It wins on cost per page.

Cohere's Parse 5, a 2.3-billion-parameter model, offers cost-effective document parsing at \$1.50 per 1000 pages, prioritizing structure over...

1 week ago



Kr 36Kr

LatentVerse Secures Hundreds of Millions Yuan Investment Led by GL Ventures, Rejects World Model Company Label to Build Embodiment-Native Brain | 36Kr Exclusive

By Deng Yongyi. Edited by Zhang Yuxin. The embodied intelligence track may no longer need an additional world model company — but a...

1 month ago



B Bessemer Venture Partners

Vision-native AI opportunities: a precursor to intelligent robotics

Next-gen computer vision models, VLMs, are ushering in ubiquitous physical intelligence.

2 Dec 2025



f finance.biggo.com

Ex-Alibaba AI Lead Lin Junyang Launches New Lab, Seeks \$2 Billion Valuation for World Model and Embodied Brain Venture

Just two months after his sudden departure from Alibaba, former Qwen lead researcher Lin Junyang is already back in the game with a new AI...

14 May 2026



VB VentureBeat

Three ways AI is learning to understand the physical world

Large language models lack grounding in physical causality — a gap world models are designed to fill. Here's how three distinct...

20 Mar 2026



P2 Pulse 2.0

Wirestock: \$23 Million Series A Raised To Scale Multimodal AI Data Platform

Wirestock announced the closing of a \$23 million Series A funding round led by Nava Ventures with participation from SBVP, Formula VC,...

14 May 2026

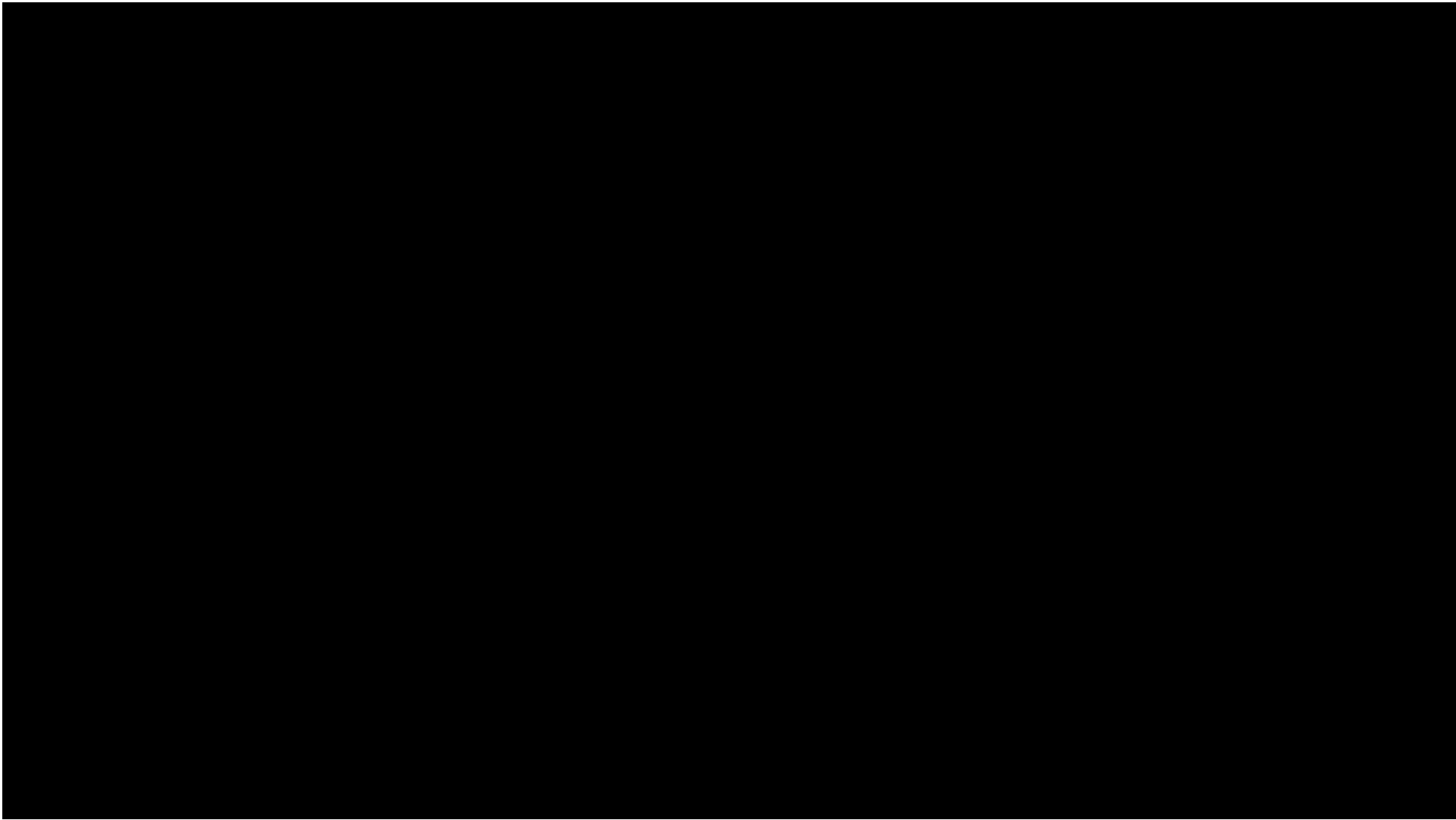


36Kr European Central Station

Exclusive information from *Intelligent Emergence* shows that embodied intelligence firm LatentVerse has completed a seed round financing of several hundred million RMB, with investors in this first round including GL Ventures, Qingliu Capital, Zhiyuan, StarMotion Era, InnoAngel and other funds.

In the past two years, VLA (Vision-Language-Action) has been the most mainstream technical route for embodied intelligence. From RT-2 to the widely popular Pi-0.5, the default evolution path of the industry is that as long as the vision-language model is connected to action output and fed with sufficient robot operation data, the model can learn to act in the physical world.

However, VLA has a fundamental limitation: it can only utilize data with action annotations. Such data is collected by operating robotic arms manually, with an average cost of more than 1000 yuan per hour, and requires dedicated operators, leading to relatively low efficiency.



Why Networking QoS Matters for a Strawberry Picking App (VLA)

A VLA robot must see, understand and act in the real world — it needs network services with low, bounded latency, high reliability, and timely return of action tokens.

VLA-powered Strawberry Picking



- Video** (high rate)
Visual scene for perception
- Audio** (real-time)
Human instructions (e.g., "pick the ripe one")
- Language**
Task goals (e.g., "pick ripe strawberries on the left")
- Sensors** (low latency)
Position, touch, force, etc.

Multimodal data streams



Timely delivery of perception, decisions and action tokens

Low & bounded latency
Keep the control loop responsive

High reliability
Avoid packet loss and interruptions

Adaptive bandwidth
Handle bursty video and model traffic

Timely return of action tokens
Ensure actions are executed without delay

Closed-loop: Perceive → Decide → Act

VLA Model (on edge/cloud)

Vision

"Find ripe strawberries"

Language

"Pick the red one on the left"

Action

"Move, grasp, pick"

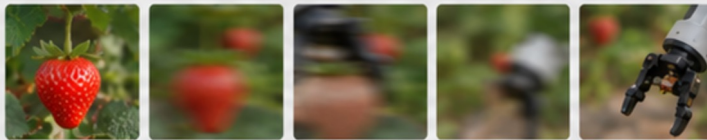


Action tokens
(e.g., motion commands, grasp)

Timely return of action tokens is critical for safe and effective control.

Without VLA-oriented QoS (Best-effort network)

- ❌ High / variable latency
- ❌ Packet loss
- ❌ Jitter
- ❌ Delayed action tokens
- ❌ Inconsistent behaviour



Delayed or incomplete perception → wrong or missed actions

With VLA-oriented QoS (Our goal)

- ✅ Low and bounded latency
- ✅ High reliability
- ✅ Adaptive bandwidth
- ✅ Timely return of action tokens
- ✅ Safe and efficient operation



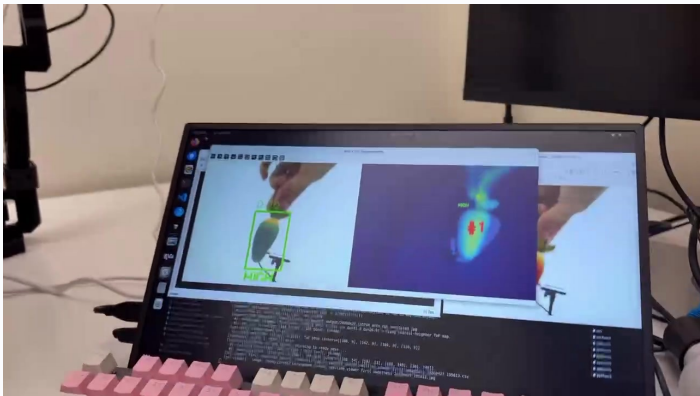
Timely, high-quality perception → correct, safe and efficient actions

Reliable and timely networking is a key enabler for real-world VLA applications.

Strawberry-Picking VLA: Why Network QoS Matters

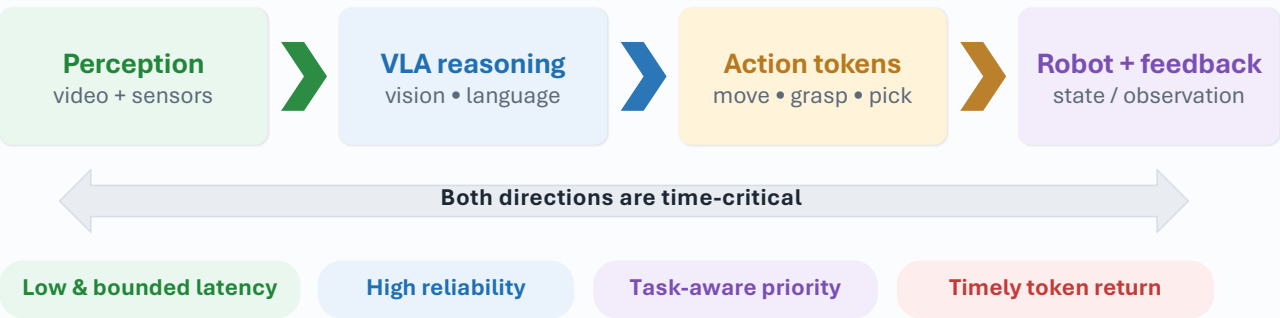
QoS must protect the whole closed loop — **perception streams**, **action-token delivery** and feedback.

VLA app & multimodal perception



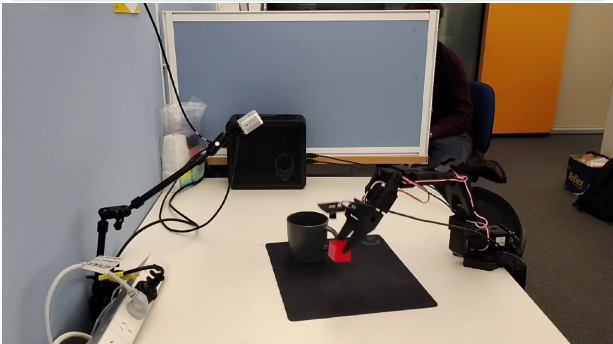
Video + sensing provide the context for VLA reasoning.

Closed-loop VLA service over the network



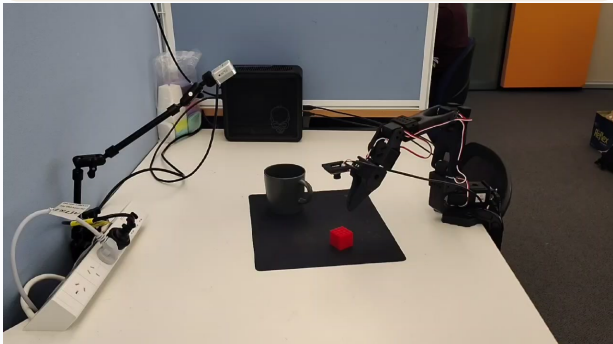
A best-effort network can break the control loop even when the VLA model is correct.

NORMAL — timely closed loop



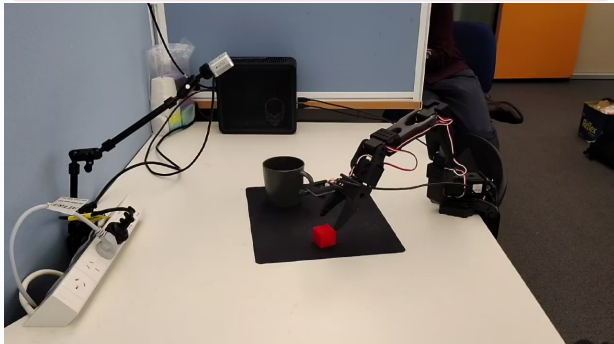
Timely perception + action tokens → confident action and a successful pick

HESITATION — delayed information



Delay or jitter in perception / action tokens → slow response or a missed action

CONFUSION — incomplete perception

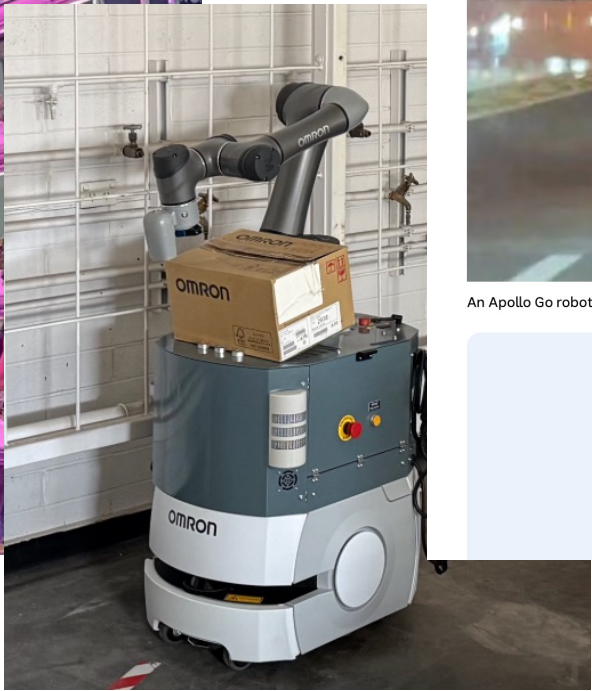


Incomplete or inconsistent perception → wrong target and failed action

VLA-oriented QoS must protect perception, action-token delivery and feedback — not just move data.

Vertical Farming (Protected Cropping):

- <https://www.youtube.com/watch?v=aLNK9FjRtA4>



Robotaxi outage in China strands passengers and causes traffic chaos

AI

Posted 2 Apr 2026 at 4:38am, updated 2 Apr 2026 at 8:01am



An Apollo Go robotaxi stopped in the middle of traffic due to a system failure in Wuhan. (Social media via Reuters)

In short:

Robotaxi passengers have been stranded in Wuhan after the driverless cars stopped due to a "system malfunction".

Local police said more than 100 vehicles were stalled, with some people stuck in the middle of busy roads.

AI CHANGES THE QUESTION

Why are you here in 2026?

AI can explain TCP, write socket code, and debug firewall rules in seconds. So why a whole term on this?

The students who thrive aren't outsourcing thinking to AI - *they're the ones who can tell when it's wrong.*

WHAT AI IS GOOD AT

- Explaining standard concepts
- Writing boilerplate code
- Summarising RFCs
- Suggesting plausible answers fast

WHERE AI GETS IT WRONG

- Real network debugging
- Reading actual packet captures
- Diagnosing latency & routing issues
- Knowing when its own answer is nonsense

Today's Agenda

- Course (non-technical) details
- Logistics: How we will roll
- What is this course about?
- Introduction to Computer Networks (course content begins)

Course Staff

- Lecturer-in-Charge: Wen Hu
- Course Admin: Isura Bamunusinghe
- Tutors (17)

Divya Prabha Tyagi
Eugene Briones
Hao YU
Hari Ganesan
Isura Bamunusinghe
Jean Du Plessis
Julian Zincone
Mahshid Gohari

Michael Zheng
Pierre Azzi
Ransika Hewa Kasakarage Gunasekara
Rushi Babaria
Tianlang Zhang
Varun Niraj Agarwal
Vinayak Kuanr
Yihe Yan
Zakia ZAMAN



You

- Mix of UG (mostly 2nd/3rd year) and PG (mostly 1st year)
 - ~700
- Mostly CSE students but a few from other Engineering schools (Mech, EET) and Faculties (Business, Science, Law)
- Assumed Knowledge:
 - COMP1927/COMP2521/MTRN3500
 - Good understanding of data structures, algorithms, basic probability theory
 - Proficient in one of the following programming languages: C, Java or Python
 - We DO NOT assume that you know anything about computer networks

Resources

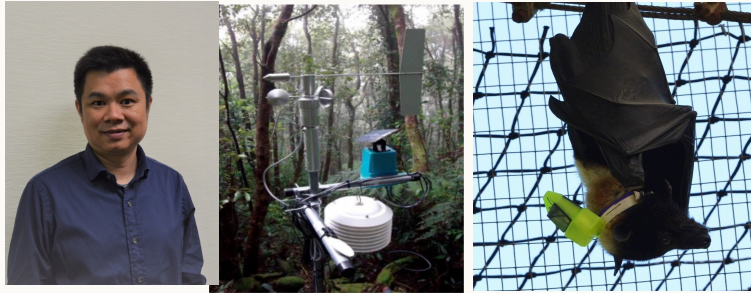
Very important



- <https://webcms3.cse.unsw.edu.au/COMP3331/26T3/>
- Everything is posted on the course website
 - **ECOS (PLEASE READ THIS THOROUGHLY)**
 - Lecture Notes
 - Video Recordings
 - Lab Schedules, Allocations and Locations
 - Assignment and Lab Exercises
 - Homework Problems
 - Exam Information
 - Consultation hours
 - **Announcement:** Your responsibility to check the announcement forum on regular basis for important updates/changes to schedule, etc.
 - **Your active participation and interaction is crucial to ensure that all of us get the most out of this course**
 - Note: You will need to login using your **zID/zPass**

WHO'S TEACHING YOU

About me



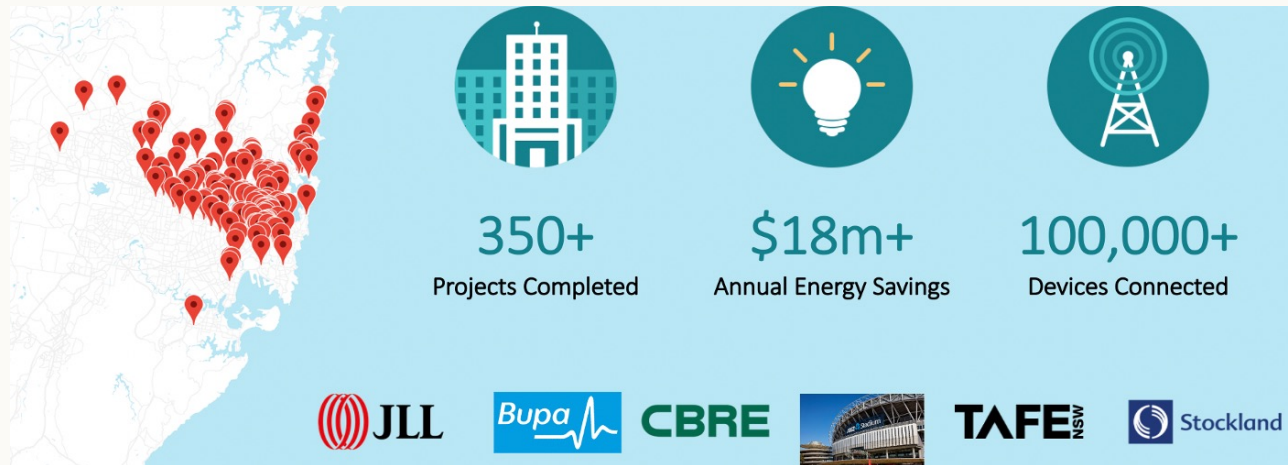
Professor of Computer Science and Engineering @ UNSW

Editor-in-Chief, ACM Transactions on Sensor Networks

20+ years of research and industry (CSIRO) experience in networking, IoT, smart buildings, smart farming and intelligent sensing

30 HDR completions, 5 patents, 200+ publications, 12,000+ citations

More about me: <https://www.cse.unsw.edu.au/~wenh>



12:02

The era of talking homes has arrived

news.com.au

Finance Real Estate

The era of talking homes has arrived

DECEMBER 27, 2018 7:57pm

Dr Wen Hu. Picture: Quentin Jones. Source: Supplied

LISA HERBERTSON | The Daily Telegraph

An Australian company, using a wireless solution developed by UNSW, is rolling out technology across the country that allows buildings to monitor themselves and report problems automatically and even talk to your smartphone.

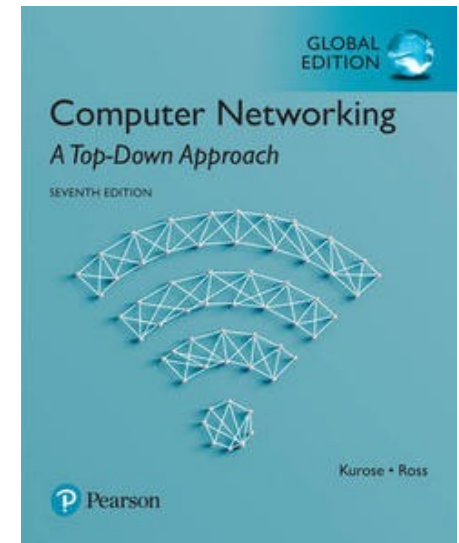
Prof Wen Hu



Course Material

- **Textbook:** *Computer Networking: A Top Down Approach*, Jim Kurose, Keith Ross, Addison-Wesley (Pearson), 8th Edition, 2025
- Lecture Notes (on WebCMS)
- Links/articles on additional material
- Reference Books:
 - *Computer Networks: A Systems Approach*, Larry Peterson and Bruce Davie, Morgan Kaufmann, 6th Edition, 2022.
 - *Unix Network Programming Volume 1 - Networking APIs: Sockets and XTI*, W. Richard Stevens, Prentice Hall, 3rd Edition, 2023
 - *Java Network Programming*, E. R. Harold, O'Reilly, 4th Edition, 2013.
 - *Computer Networks*, Andrew Tanenbaum and David Wetherall, 6th Edition, Pearson, 2021.

Links to programming help



Course Aims

- To gain in-depth introduction to the key topics in the field of computer networks, including the Internet
- To obtain hands-on understanding of networking protocols
- To gain skills in network programming, designing and implementing network protocols, evaluating network performance and problem solving
- To build necessary foundational knowledge required in more advanced networking courses

Teaching/Learning Strategies

- Lectures (8 x 4 hours and 1 x 2 hours)
- Labs
 - Hands-on learning
- Programming Assignment
 - Network programming and protocol design
- Weekly Homework (Self-assessed)
 - Problem solving skills

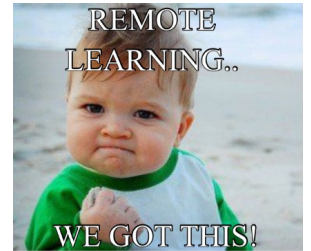
Lectures

- In-person face-to-face: Monday 11:00 – 13:00 and Thursday 09:00 - 11:00
- Weeks 1-5 and 7-10
 - 2 x 2-hour lectures x 8 weeks + 1 x 2-hour lecture x 1 week
 - Public holiday: 05/10 (Week 4, Monday)
- We will focus on most important concepts and supplement with
 - Problem solving exercises
 - Discussions
- Certain material will be left for self study
 - These will be indicated on the lecture notes
- In-lecture polls and quizzes
 - For you to reinforce concepts
 - For me to get an indication of your understanding

Labs

- 2-hour lab sessions starting **Week 2** (Weeks 2-5, 7-10)
- **In-person or On-line labs**
- Hands-on experiments related to concepts covered in lectures
 - Wireshark packet sniffer, ns-3 network simulator, other network measurement tools, socket programming practice
- 8 lab sessions:
 - 5 Lab Exercises (guided by tutors)
 - Lab report to be submitted
 - Highly encouraged to attempt lab tasks before attending labs
 - 4 Problem-based learning sessions (Tutorials in Weeks 3, 5, 7& 10)
 - No marks

Online Labs: VLAB



- Access CSE lab environment on your own machine remotely
- Uses VNC
- Recommended client: TigerVNC (<https://tigervnc.org>)
- Quick start:
https://taggi.cse.unsw.edu.au/FAQ/Really_quick_guide_to_VLAB/
- UNSW VPN: <https://www.myit.unsw.edu.au/>
- You will need to know basic command line Linux commands:
<http://www.unixguide.net/linux/linuxshortcuts.shtml>



Getting help

- Use online discussion forums on Discourse (Join here: <https://discourse03.cse.unsw.edu.au/26T3/COMP3331>)
 - Fellow students benefit from your questions
 - Fellow students can answer your questions
 - Develop a community
- Use cs3331@cse.unsw.edu.au for communication with us.
 - DO NOT email LiC/admin on personal email address
- Consultation hours
 - LiC for lecture-related help – 1 hour each week
 - Tutor Consultations for assignment help - C/Python/Java
- Tutors
 - Establish an agreeable mode of communicating with your tutor

Revisions based on myExperience Feedback

- Student feedback from Terms 1, 2 was generally positive
- Programming specific lab during Weeks 3, 7 focusing on socket programming and how to work with multi-threading
- Assignment
 - Specs will be more concise and released before Week 3
 - More consultations sessions through Weeks 4-9

Code of Conduct

- CSE offers an inclusive learning environment for all students. In anything connected to UNSW, including social media, these things are student misconduct and will not be tolerated:
 - racist/sexist/offensive language or images
 - sexually inappropriate behaviour
 - bullying, harassing or aggressive behaviour
 - invasion of privacy
- Show respect to your fellow students and course staff
- Staff are also reminded to show respect to students

Plagiarism

What is plagiarism?

Presenting the (thoughts or) work of another as your own. Cheating of any kind constitutes academic misconduct and carries a range of penalties. Please read course intro for details.



Examples of inappropriate conduct:

- groupwork on assignments/labs (discussion OK)
- allowing another student to copy your work
- getting your hacker cousin to code for you
- purchasing a solution to the assignment

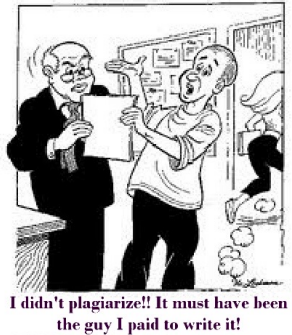
Remember: You are only cheating yourself and chances are you will get caught!

Using Artificial Intelligence at UNSW:

<https://www.unsw.edu.au/student/managing-your-studies/academic-skills-support/toolkit/ai>

Plagiarism

- Labs, assignments, exams must be entirely your own work
- You **can not** work on assignment as a pair (or group)
- Plagiarism will be checked for and penalized
- Plagiarism may result in suspension from UNSW
- Scholarship students may lose scholarship
- International students may lose visa
- Supplying your work to any another person may result in loss of all your marks for the lab/assignment
- If you store your code in online repositories DO NOT MAKE IT PUBLICLY ACCESSIBLE (THIS IS ASSUMED TO BE PLAGIARISM)



Assessment

- **Hands-on – 40%**
 - Labs 20%
 - Assignment 20%
 - Assignment released before Week 3, due in Week 9
 - Implement a networked application or protocol
 - We assume you are proficient in one of C/Java/Python (coding skills are must in most practical networking jobs!)
- **Concepts and theory – 60%**
 - Mid-term test (20%):
 - **Week 7 (Monday, 11am to 1pm).**
 - Close-book online exam (**Inspera ---On campus and Invigilated**)
 - Final Exam (40%)
 - End of term
 - Close-book online exam (**Inspera --- On campus and Invigilated**)
 - Hurdle – **must score at least 40% to pass the course**
 - **Inspera platform**

Assessment

NOTE: To pass the course, a student MUST receive at least **40% marks on the final exam**

```
lab = marks for lab exercises (20 marks)
assign = mark for the programming assignment (20 marks)
midTerm = mark for the mid-semester exam (20 marks)
scaledfinalExam = scaled mark for the final exam (out of
40 marks)
mark = lab + assign + midTerm + scaledfinalExam
Grade:
= HD|DN|CR|PS if mark >= 50 && scaledfinalExam >= 16
= FL          if mark < 50 || scaledfinalExam < 16
```

NOTE: If you cannot clear the final exam hurdle (after scaling), reported grade would be 'UF' with maximum marks reported as 40

2026T2:

- Labs: ~90%
- Assignment: ~80%
- ~9% UF

How to do well in this course

- Keep up with and absorb all the content
 - Clear weekly tasks; do not accumulate
 - **This is an intense course requiring full attention**
- A critical/analytical viewpoint will help
- Solve all homework/practice problems
- Do the lab exercises *yourself*
- Do the assignment *yourself*
- Practice, practice, practice



Online/Hybrid Delivery

- We all need to work together
- Course Staff
 - Regular communication about upcoming deadlines (weekly notices)
 - Timely response to questions
 - Timely feedback on assessments
- Students
 - Take responsibility
 - Be aware of deadlines/deliverables and how to access resources
 - Links for lectures/labs/consults/exams
 - VLAB for labs and assignments
 - Check course notices regularly
 - Ask questions through the appropriate channels (online forum is preferred)
 - Participate in lectures and forum (community building)



What is this course about ?

- Introductory course in computer network
- Learn *principles* and *practice* of computer networking
- We use the **Internet** as a vehicle to understand the core concepts of networking

What is this course about ?

1. To learn how the Internet works
 - Internet is a complex global infrastructure
 - What are the organising principles behind the Internet?
 - What really happens when you “browse the Web”?
 - What are TCP/IP, DNS, HTTP, BGP, NAT, VPNs, 802.11,..... anyway?



What is this course about ?

1. To learn how the Internet works

- Internet is a complex global infrastructure
- What are the organising principles?
- What really happens when you “browse”?
- What are TCP/IP, DNS, HTTP, BG



2. To learn the fundamentals of computer networks

- What issue you need to take into consideration to make a computer network work well?
- What design strategies have proven valuable?
- How do we evaluate network performance?

Where could I go from here?

- COMP 4336/9336: Mobile Data Networking
- COMP6733: Internet of Things Design Studio
- COMP 9334: System Capacity and Planning
- COMP 3441/9441: Security Engineering
- COMP 4337/9337: Securing Wireless Networks
- COMP 9333: Advanced Computer Networks (Refreshed)

- Thesis/Coursework Projects
- Research Degree (MPhil, PhD)

